Measurement – Mass and Volume Towards Density

DATA ANALYSIS

Table 1: Volumes and Density Regular 3D Shapes

	Volume Equation (in terms of a, b, c)	**Calculated Volume** (mm³)	**Calculated Volume** (cm³)	**Calculated Density** (g/cm³)	**Deviation (g/cm³)** $\lvert(\overline{X} - X_i)\rvert$
(1) Cube					
(2) Cuboid					
(3) Triangular Prism					
(4) Cylinder					
(5) Square Pyramid					
(6) Triangular Pyramid					
(7) Cone					
(8) Ovoid					
(9) Ellipsoid					
(10) Sphere					

Using of the Densities of the Regular 3D Shapes, Calculate the Mean (Average) Density of Beech Wood

Using the Mean Density of Beech Wood and the Mass of Your Ovoid, Calculate the Volume of the Ovoid

Calculate the Mean (Average) Deviation of the Density of Beech Wood

Calculate the Relative Deviation (in %)

Calculate the Percent Error (in %)

Calculate the Percent Error (in ppt)

Table 2: Measurements of Fluorite

	Calculated Density (g/mL)	**Deviation (g/cm³) $\lvert(\overline{X} - X_i)\rvert$**		**Calculated Density (g/mL)**	**Deviation (g/cm³) $\lvert(\overline{X} - X_i)\rvert$**
1			11		
2			12		
3			13		
4			14		
5			15		
6			16		
7			17		
8			18		
9			19		
10			20		

Using of the Densities of Fluorite, Calculate the Mean (Average) Density of Fluorite

Calculate the Mean (Average) Deviation of the Density of Fluorite

Calculate the Relative Deviation (in %)

Calculate the Percent Error (in %)

Calculate the Percent Error (in ppt)